ModelIt!

Teacher's Guide

Earth's Hidden Water

97% Is Undrinkable, 2% Is Frozen. We Live on the Rest.

Quick Reference

Grade Level	Grade 5
NGSS Standard	5-ESS2-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 5-ESS2-2

Describe and graph the amounts of water and fresh water in various reservoirs to provide evidence about the distribution of water on Earth.

Three-Dimensional Learning

Science Practice	**Analyzing and Interpreting Data** — Students analyze how different conditions affect water distribution.
Disciplinary Core Idea	**ESS2.C The Roles of Water in Earth's Processes** — Water cycles through Earth's systems in various forms.
Crosscutting Concept	**Systems and System Models** — Students model the freshwater system with competing inputs and outputs.

CAST Testing Alignment

MULTIPLE CHOICE:

During a heat wave with no rain, what happens to the amount of fresh water available in a reservoir?

CONSTRUCTED RESPONSE:

Explain why high temperatures AND low precipitation together create severe water shortages.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

Earth is a water planet, but most of that water isn't usable. Understanding water distribution is essential for managing this critical resource.

Water Distribution

97% of Earth's water is saltwater in oceans. 2% is frozen in glaciers and ice caps. Less than 1% is accessible fresh water in rivers, lakes, and groundwater.

The Water Cycle

Water moves continuously between reservoirs through evaporation, condensation, and precipitation. This cycle is powered by solar energy.

Evaporation Factors

Higher temperatures increase evaporation rates. Hot, dry conditions can cause water to evaporate faster than precipitation can replace it.

Real-World Crisis

In 2018, Cape Town, South Africa nearly ran out of water during a severe drought. They called it 'Day Zero' - the day taps would be turned off.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — Locate the System
Students identify precipitation, temperature, freshwater availability, and evaporation as system components.

E — Establish Relationships
Students establish that precipitation increases freshwater while evaporation decreases it - a key negative relationship.

V — Visualize & Model
Students build the water balance model showing competing forces.

E — Evaluate Outcomes
Students run drought scenarios and observe freshwater availability crashing.

R — Revise & Extend
Students add human water use to see how human activity affects the system.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"97% of Earth's water is undrinkable. 2% is frozen."*

Watch For

Let that statistic sink in.

Slide 2: Learning Objectives

2 min

Say: *"Today we model what controls freshwater availability."*

Watch For

Define key vocabulary.

Slide 3: Big Question

2 min

Say: *"What determines if your city has enough water?"*

Watch For

Discuss local water sources.

Slide 4: LEVER Framework

2 min

Say: *"We'll build a model of the water balance."*

Watch For

Preview the activities.

Slide 5: Activity 1: Components

8 min

Say: *"Let's identify inputs and outputs of the water system."*

Watch For

Emphasize the competing forces.

Slide 6: Activity 2: Connections

8 min

Say: *"Notice: evaporation has a NEGATIVE effect on freshwater."*

Watch For

Ensure students understand why.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's simulate the Cape Town water crisis."*

Watch For

Students run drought + heat scenario.

Slide 8: Discoveries

5 min

Say: *"This is why water managers worry about droughts."*

Watch For

Connect to student experiences with water.

Slide 9: STEM Challenge

10 min

Say: *"Design a plan to save a city's water supply."*

Watch For

Form groups and distribute materials.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Precipitation Amount, Temperature
- **INTERNAL:** Freshwater Availability, Evaporation Rate

Relationships

- **Precipitation to Freshwater Availability:** POSITIVE (+) — More rain/snow adds more water to rivers, lakes, and groundwater.
- **Temperature to Evaporation Rate:** POSITIVE (+) — Higher temperatures cause water to evaporate faster.
- **Evaporation Rate to Freshwater Availability:** NEGATIVE (-) — More evaporation removes water before communities can use it.

Key Discoveries

- Less than 1% of Earth's water is accessible fresh water
- Precipitation ADDS to freshwater availability
- Evaporation REMOVES from freshwater availability
- Hot, dry conditions create water crises for communities

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"There's plenty of water everywhere"	Less than 1% of Earth's water is accessible fresh water - and it's not evenly distributed.
"Rain refills everything quickly"	Groundwater can take centuries to recharge. Aquifers being depleted now may not refill in our lifetimes.
"Water disappears when it evaporates"	Water cycles back eventually, but not necessarily to the same place. It might rain far away.